EXPERIMENT 22: Prolog Program to Print a Particular Car by Fan or Not

Aim

To write and execute a Prolog program to represent car-related facts and determine whether a particular car satisfies the required condition using suitable Prolog queries.

Algorithm

1. Start the Prolog program.
2. Define facts containing the required car and related information.
3. Define the required relationship or condition.
4. Create a rule to determine whether the specified car satisfies the condition.
5. Query the knowledge base for a particular car.
6. Display whether the required condition is true or false.
7. Stop.

Program Code:

```
% Car database


car(bmw).

car(audi).

car(toyota).

car(honda).

car(ford).


% Check whether car is present

check_car(Car) :-

    car(Car),

    write(Car),

    write(' is available.'), nl.
```

```
check_car(Car) :-

    \+ car(Car),

    write(Car),

    write(' is not available.'), nl.
```

Output:

```
(125 ms) yes
| ?- consult('C:/Users/Hp/Documents/ex-22.pl').
compiling C:/Users/Hp/Documents/ex-22.pl for byte code...
C:/Users/Hp/Documents/ex-22.pl compiled, 17 lines read - 1195 bytes written, 19 ms

yes
| ?- check_car(bmw).
bmw is available.

true ? |
```

Result

The Prolog program was successfully implemented to represent the required car information and determine the specified condition using Prolog queries.

Note: The uploaded image's wording for Experiment 22 is unclear ("particular car by fan or not"). I have kept the answer general rather than inventing a different problem.